euispice_coreg package presentation for EUI

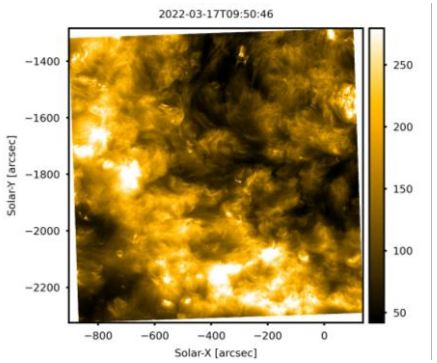
euispice_coreg is an open source python package to correct the pointing information in the FITS files header for EUI or SPICE

Main points:

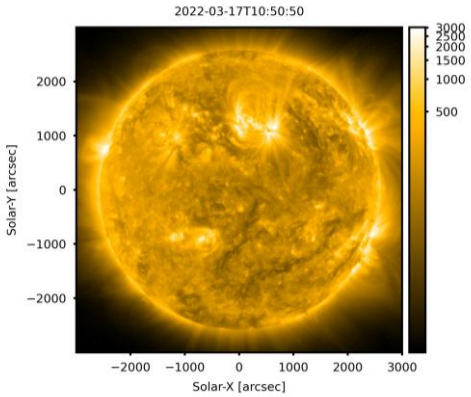
- Available in https://github.com/adolliou/euispice_coreg (with pip install git+) and as a package in pypi. Described in Appendix A of Dolliou *et al,* A&A, 2024. Inspired from the co-alignment technique used in Berghmans *et al.*, A&A, 2021.
- Cross-correlates images by shifting FITS header values (CRVAL1, CRVAL2, CROT, CDELT1, CDELT2)
- Can perform the reprojection in **Helioprojective** or in **Carrington** coordinates.
- Parallelized and fast. The code is adapted to large datasets.
- Includes a jitter correction tool.
- Easy to use and includes detailed instructions in the github page.
- Can be used to any FITS files with a similar format to EUI. I used it to co-align IRIS/SJI 1400 with AIA 1700 images.

Example : co-align HRIEUV with AIA 171 in carrington coordinates

HRIEUV



AIA 171 (1 hour later)



Repeat for all of the lag arrays

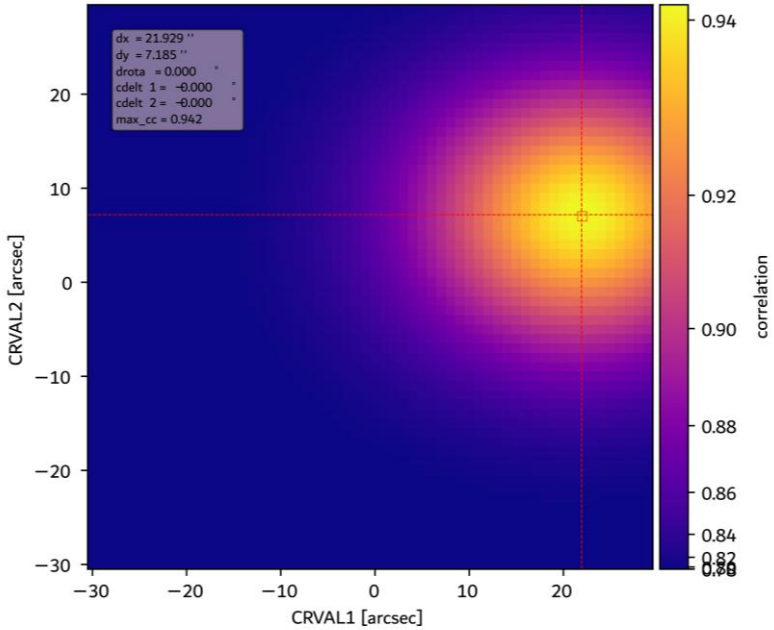
Shift HEADER values

CRVAL1/2 CDELT1/2 CROTA

Transform into Carrington grid provided by user

Cross-correlation

Apply best (interpolated) shift to header keywords



Example : co-align HRIEUV with AIA 171 in carrington coordinates

Example of code :

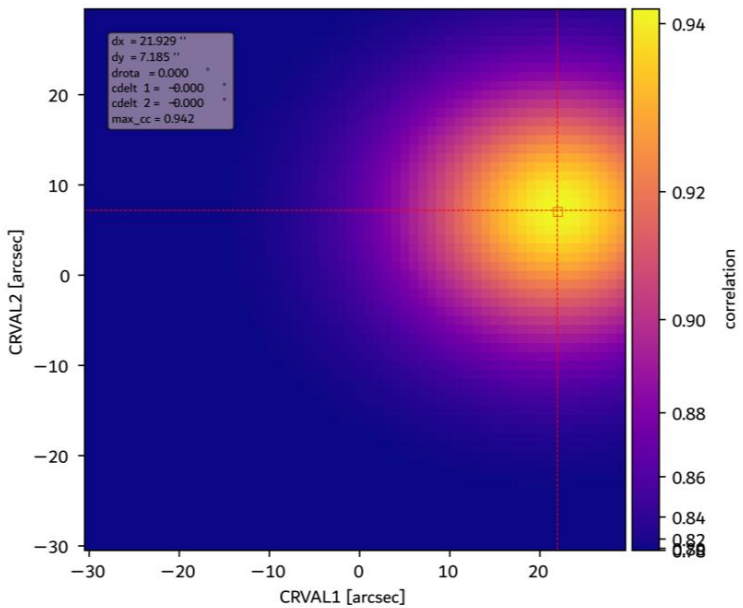
```
from euispice_coreg.hdrshift import Alignment
import numpy as np

if __name__ == "__main__":

    path_fsi =
("https://www.sidc.be/EUI/data/releases/202204_release_5.0/L2/2022/03/17/solo_L2_eui-fsi174"
                "-image_20220317T095045281_V01.fits")
    path_hri =
("https://www.sidc.be/EUI/data/releases/202204_release_5.0/L2/2022/03/17/solo_L2_eui-hrieuv174"
                "-image_20220317T105045276_V01.fits")


   # Create the Carrington grid

    lonlims = (230, 260)
    latlims = (-65, -35)
    shape = (2400, 2400)

    # Create the lag parameters in the FITS headers

    lag_parameters = {
        "lag_crval1": np.arange(-30, 30, 1),
        "lag_crval2": np.arange(-30, 30, 1),
        "lag_crota":  np.arange(-1.0, 1.0, 0.5),
        "lag_cdelt1": [0],
        "lag_cdelt2": [0],
    }
```

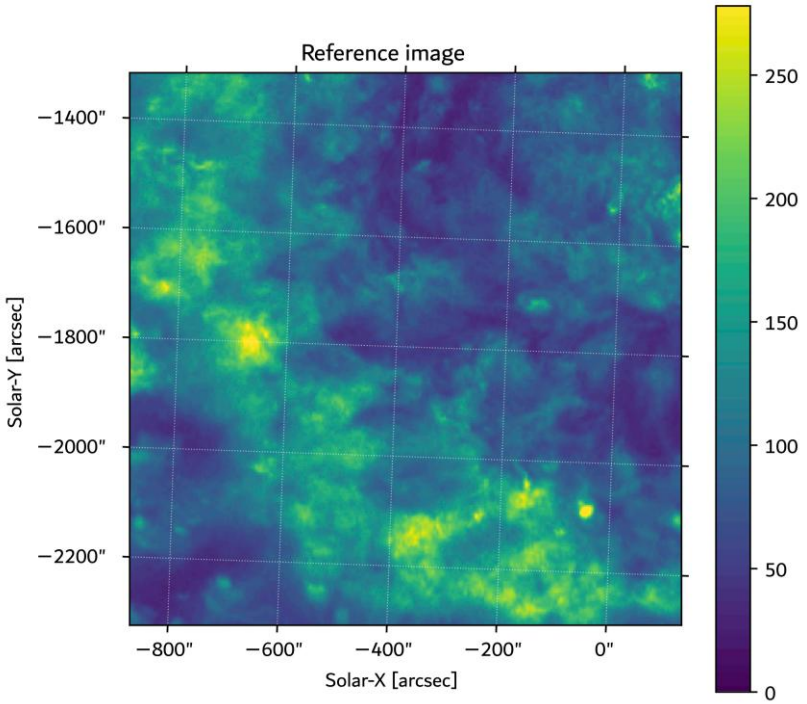
```
    parallelism = True
    counts_cpu_max = 8


    A = Alignment(
            large_fov_known_pointing  = path_fsi,
            small_fov_to_correct      = path_hri,
            large_fov_window          = -1,
            small_fov_window          = -1,
            parallelism               = parallelism,
            counts_cpu_max            = counts_cpu_max,
            **lag_parameters
                        )
    results = A.align_using_carrington(
            shape   = shape,
            lonlims = lonlims,
            latlims = latlims,
            )
```

Example : co-align HRIEUV with AIA 171 using carrington coordinates

The package also includes :

- Plotting tools for the correlation and the co-alignment results.
- Function to save the new co-aligned FITS files.





Important points :

- **Only the values in the header keywords are changed in the co-aligned FITS files, not the data.**
- If co-alignment is done in Carrington coordinates, **co-aligned FITS files can then be used with sunpy and reproject function.**